Homework 4

MATH 518 - Fall 2025

Due on Wednesday, November 12, 11:59pm on Crowdmark

Total of points = 30 points

Please submit each question separately on Crowdmark. Unless stated otherwise, k is an algebraically closed field of characteristic zero.

1. (10 points) Let X, Y be quasi-projective varieties, and $f\colon X \to Y$ a map. Suppose that $X = \bigcup_{i\in I} U_i$ and $Y = \bigcup_{i\in I} V_i$ are two covers of X and Y by open subsets $U_i \subseteq X$ and $V_i \subseteq Y$, such that $f(U_i) \subseteq V_i$ for all $i \in I$.
   (a) Show that f is a morphism if and only if each restriction $f_i \coloneqq f|_{U_i}\colon U_i \longrightarrow V_i$ is a morphism.
   (b) If U_i and V_i are affine varieties, show that $f_i\colon U_i \to V_i$ is a morphism if and only $f_i^*(k[V_i]) \subseteq k[U_i]$.

2. (10 points) Let $f\colon X \longrightarrow Y$ be a morphism of quasi-projective varieties. Let $X' \subseteq X$ and $Y' \subseteq Y$ be closed subvarieties such that $f(X') \subseteq Y'$. Show that the restriction $f|_{X'}\colon X' \longrightarrow Y'$ is a morphism.
   Hint: Show it for affine varieties first, then use the previous exercise

3. (10 points) Let $X = k^2 \smallsetminus (0,0)$.
   (a) Show that $k[X] = k[x,y]$.
   (b) Deduce that X is not isomorphic to an affine variety.
   Hint: Use the fact that if Y is an affine variety, then any proper ideal $I \subset k[Y]$ defines a nonempty set $V(I)$, by the Nullstellensatz.
   (c) Deduce that the union of two open affine subvarieties of a variety is not necessarily affine.